

Haochen Sun

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University of Waterloo, 200 University Avenue West, Waterloo, Ontario, N2L 3G1, Canada

EDUCATION

University of Waterloo

Sep 2022 – Present

PhD in Computer Science

Waterloo, Canada

- Supervisor: Prof. Xi He
- Research focus: data privacy and machine learning security.
- Honors and awards: **SIGMOD 2026 Best Paper Award** for GPM; David R. Cheriton Graduate Scholarship.

The Hong Kong University of Science and Technology (HKUST)

Sep 2018 – Jul 2022

BSc in Data Science and Technology, and in Computer Science

Hong Kong, China

- GPA: 3.888/4.3; major GPA: 4.041/4.3.
- Honors and awards: Chern Class Scholarship (Department of Mathematics), Zhiyuan Scholarship (China Soong Ching Ling Foundation), HKUST Scholarship Scheme for Continuing Undergraduates, and Dean's List.

RESEARCH EXPERIENCE

Faithfulness of Differential Privacy Mechanism Execution

May 2024 – Present

PhD Research Project (with Prof. Xi He)

University of Waterloo

- Investigating vulnerabilities from unfaithful execution of differential privacy mechanisms, focusing on additional privacy leakage incurred and defenses to ensure execution integrity.
- Our GPM work received the **SIGMOD 2026 Best Paper Award**.

Zero-Knowledge Deep Learning

Sep 2022 – Apr 2024

University of Waterloo

- Specialized zero-knowledge proof (ZKP) protocols for deep learning with CUDA implementations.
- The first working ZKP scheme for 13B-parameter LLMs and for training 10M-parameter neural networks.

Air Pollution Forecasting with Deep Learning

Jan 2020 – Nov 2022

Undergraduate Research and RA (with Profs. Jimmy Fung and Xingcheng Lu)

HKUST

- Designed deep learning architectures for modeling the spatio-temporal distribution of air pollutants.
- Improved regional air pollution forecasting accuracy by 30%.

PUBLICATIONS

Only (co-)first-author publications are listed below.

1. **(SIGMOD 2026 Best Paper Award)** [Haochen Sun](#) and Xi He. “GPM: The Gaussian Pancake Mechanism for Planting Undetectable Backdoors in Differential Privacy.” *Proceedings of the ACM on Management of Data (PACMMOD)*, 2026.
2. [Haochen Sun](#) and Xi He. “VDDP: Verifiable Distributed Differential Privacy under the Client–Server–Verifier Setup.” *Theory and Practice of Differential Privacy (TPDP)*, 2026.
3. Shufan Zhang*, [Haochen Sun](#)*, Karl Knopf*, Shubhankar Mohapatra, Wei Pang, Calvin Wang, Yingke Wang, Masoumeh Shafieinejad, David Emerson, and Xi He. “FedDPSyn: Federated Tabular Data Synthesis with Computational Differential Privacy.” *Theory and Practice of Differential Privacy (TPDP)*, 2025.
4. [Haochen Sun](#), Jason Li, and Hongyang Zhang. “zkLLM: Zero-Knowledge Proofs for Large Language Models.” *ACM Conference on Computer and Communications Security (CCS)*, 2024.
5. [Haochen Sun](#), Tonghe Bai, Jason Li, and Hongyang Zhang. “zkDL: Efficient Zero-Knowledge Proofs of Deep Learning Training.” *IEEE Transactions on Information Forensics and Security (TIFS)*, 2024.
6. [Haochen Sun](#), Jimmy C. H. Fung, Yiang Chen, Zhenning Li, Dehao Yuan, Wanying Chen, and Xingcheng Lu. “Development of an LSTM broadcasting deep-learning framework for regional air pollution forecast improvement.” *Geoscientific Model Development (GMD)*, 2022.
7. [Haochen Sun](#), Jimmy C. H. Fung, Yiang Chen, Wanying Chen, Zhenning Li, Yeqi Huang, Changqing Lin, Mingyun Hu, and Xingcheng Lu. “Improvement of PM_{2.5} and O₃ forecasting by integration of 3D numerical simulation with deep learning techniques.” *Sustainable Cities and Society (SCS)*, 2021.

ACADEMIC SERVICES

- **(Sub)Reviewer:** Conferences: CCS, SIGMOD, VLDB, NeurIPS, SaTML, AISTATS; Journals: TDSC, TMLR, TKDE.
- **Teaching Assistant, University of Waterloo:** Courses: CS 116, CS 135, CS 246, CS 330, CS 480.